

Post-Recombination Registration and the Bright-End JWST Galaxy Excess

A Flux/MIP Collapse-Gate Model

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Abstract

The abundance of bright galaxies observed by JWST at $z \geq 10$ has renewed interest in mechanisms that can enhance early massive-halo formation without disrupting the tightly constrained CMB acoustic scale, reionization optical depth, or late-time structure growth. We introduce a Flux/MIP registration framework in which nonlinear collapse is treated not only as a gravitational threshold crossing, but as a substrate-level registration event: an overdense region becomes a persistent classical bound structure only when its state can be recorded faster than environmental scattering erases it.

We identify recombination as a global transition from a scattering-dominant to a registration-dominant medium, described by the crossing condition $\Gamma_{\text{reg}}(z) > \Gamma_{\text{loss}}(z)$. This transition defines a post-recombination memory kernel, $S_{\text{collapse}}(z, x)$, which slightly lowers the effective collapse threshold without requiring a large modification to the linear power spectrum. Using a Press-Schechter/Sheth-Tormen diagnostic pipeline, we show that a lock-in amplitude of order $A_{\text{lock}} \sim 10^{-2}$ selectively enhances the rare high-mass halo tail and can reduce the star-formation efficiency required to match bright high-redshift UV luminosity function data.

Finally, we present forecasts for Roman wide-field surveys. Unlike a uniform increase in star-formation efficiency, the registration model predicts excess clustering, enhanced field-to-field variance, and possible scale-dependent structure in the $z > 10$ bright-galaxy population. These spatial observables provide a direct falsification test of the Flux/MIP lock-in hypothesis.

Keywords: Flux cosmology; recombination; JWST; UV luminosity function; nonlinear collapse; information registration; Roman Space Telescope.

1. Introduction

The Λ CDM model remains the most successful cosmological framework for describing the CMB, large-scale structure, and late-time expansion. However, several observational tensions motivate testing extensions that remain close to Λ CDM in the background while altering the nonlinear pathway by which structure becomes observable.

One such pressure point is the early galaxy population revealed by JWST. Multiple JWST surveys have measured the rest-frame UV luminosity function at $z \gtrsim 9$, with some analyses finding unexpectedly abundant bright galaxies relative to many theoretical models, especially at $z \gtrsim 12$.

At the same time, proposed solutions must avoid violating the CMB acoustic scale, the optical depth to reionization, and low-redshift growth constraints. A mechanism that simply increases early structure everywhere risks overproducing stars, reionizing too early, or shifting the matter power spectrum.

This paper explores a different route. We propose that the relevant modification is not a large change in gravity or expansion, but a small change in the registration condition for nonlinear collapse. In the Flux/MIP framework, a bound structure is not merely a density enhancement. It is a persistent classical record. Collapse therefore requires both mass overdensity and registration stability.

The central claim is that recombination created a global change in the universe's ability to preserve information. This change leaves a delayed imprint on nonlinear collapse, selectively assisting rare high- v peaks without broadly altering the background or the linear power spectrum.

2. MIP Registration as a Collapse Condition

We interpret the cosmological substrate as an informational network composed of Minimum Intersection Points, or MIPs. In this language, a MIP is the smallest effective interaction interface at which quantum information can be accepted, rejected, scattered, or registered into a persistent classical state.

This does not require replacing standard local physics. Instead, it reframes the emergence of classical structure as a registration process. A local overdensity must not only exceed a gravitational threshold; it must also become stable against environmental erasure.

We define the environmental loss rate as the Thomson scattering rate,

$$\Gamma_{\text{loss}}(z) = n_e(z) \sigma_T c$$

where $n_e(z)$ is the free-electron density, σ_T is the Thomson cross-section, and c is the speed of light.

We model the registration rate as

$$\Gamma_{\text{reg}}(z) = A_{\text{reg}} H(z) C_\sigma(z) + B_{\text{reg}} R_\sigma(z)$$

where $C_\sigma(z)$ is the coherence fraction of the medium and $R_\sigma(z)$ is the active transition shell associated with the rapid change in coherence during recombination.

The registration index is

$$I(z) = \Gamma_{\text{reg}}(z) - \Gamma_{\text{loss}}(z)$$

Before recombination, $I(z) < 0$. The universe is scattering-dominant. Local state information is erased faster than it can be globally preserved.

At recombination,

$$I(z_*) = 0, \quad z_* \approx 1089$$

After this point, $I(z) > 0$. The universe becomes transparent, but more importantly for this framework, it becomes registrable. Thus, recombination is interpreted not only as photon decoupling, but as the first global transition from an erasure-dominant medium to a registration-dominant medium.

3. The Post-Recombination Collapse Gate

In standard spherical collapse, nonlinear structure formation is controlled by the threshold $\delta_c \approx 1.686$. In the Flux/MIP framework, this threshold is interpreted as a collapse-registration threshold. An overdense region becomes a persistent classical bound object when its gravitational tendency to collapse and its informational ability to register exceed the environmental loss rate.

We therefore introduce an effective threshold,

$$\delta_{c,\text{eff}}(z,x) = \delta_c [1 - A_{\text{lock}} S_{\text{collapse}}(z,x)]$$

where A_{lock} is the strength of post-recombination MIP lock-in assistance.

For forecasting and simulation, we decompose the memory kernel as

$$S_{\text{collapse}}(z,x) = S_0(z) [1 + \epsilon_{\chi} \chi(x)]$$

$$0 \leq S_0(z)[1 + \epsilon_{\chi} \chi(x)] \leq 1$$

$$\delta_{c,\text{eff}}(z,x) = \delta_c \{1 - A_{\text{lock}} S_0(z)[1 + \epsilon_{\chi} \chi(x)]\}$$

Here, $S_0(z)$ is the redshift-dependent delayed memory of the recombination registration shell, and $\chi(x)$ is the spatial coherence field.

The key point is that this mechanism does not primarily alter the linear growth law. It alters the condition under which rare overdensities become registered as bound halos.

This distinction matters. A direct growth modification, $\mu_{\text{eff}}(z) = 1 + A_{\mu} S(z)$, would immediately face strong CMB, lensing, and growth constraints. The collapse-gate mechanism instead acts nonlinearly and selectively, primarily in the exponential rare tail of the halo mass function.

4. Rare-Tail Amplification

The halo abundance depends strongly on the peak-height parameter,

$$v = \delta_c / \sigma(M,z)$$

For common low-mass halos, $v \leq 1$, so a small change in δ_c has little effect. For rare massive halos at high redshift, $v \gg 1$, and the mass function becomes exponentially sensitive to the collapse threshold.

To first order, the abundance ratio behaves like

$$n_{\text{Flux}} / n_{\text{LCDM}} \sim \exp[(\delta_c^2 - \delta_{c,\text{eff}}^2) / (2 \sigma^2(M,z))]$$

Thus, a small fractional reduction in the registration threshold can produce a large enhancement only where the halo population is already exponentially suppressed.

This is the central observational advantage of the model: small global effect, large bright-tail response. This explains why the effect can help with bright JWST galaxies while remaining quiet in reionization and total stellar-mass-density checks.

5. UV Luminosity Function Diagnostics

We tested the collapse-gate mechanism using a Press-Schechter and Sheth-Tormen style UVLF pipeline. The model maps halos to UV luminosities using a star-formation efficiency parameter f_{star} , then compares the predicted UV luminosity function against high-redshift JWST measurements.

The key comparison is between two explanations:

Model A: Astrophysical Brightening

$$A_{\text{lock}} = 0$$

with larger f_{star} . This model makes galaxies brighter at fixed halo abundance.

Model B: Registration-Assisted Collapse

$$A_{\text{lock}} > 0$$

with moderate f_{star} . This model changes the abundance of rare halos before the luminosity mapping is applied.

Using the real-data ingestion pipeline built from JWST UVLF measurements, we find that $A_{\text{lock}} \sim 10^{-2}$ produces the correct qualitative response: the bright end is lifted more strongly than the faint end. This is the expected signature of a collapse-threshold mechanism rather than a uniform luminosity boost.

However, the current real-data comparison should be treated as a diagnostic, not yet a final likelihood result. The correct current claim is:

$A_{\text{lock}} \sim 10^{-2}$ provides a physically motivated rare-tail lift that can reduce the need for extreme early astrophysics, especially at the bright end.

It is not yet correct to claim a definitive solution of the JWST UVLF tension until a full observational likelihood is performed across multiple datasets, systematics, selection functions, dust assumptions, scatter, and duty cycle.

6. Reionization and Stellar-Mass-Density Consistency

A successful early-galaxy mechanism must not overproduce the global stellar budget or reionize the universe too early.

The Flux/MIP collapse gate passes the first internal consistency check because its effect is rare-tail selective. It boosts the abundance of bright objects without proportionally increasing the total number of all star-forming halos.

In the v6 reionization diagnostic, the model remained compatible with a Planck-like optical-depth target for moderate A_{lock} . In the v8 stellar-mass-density diagnostic, the best-fit Flux configurations remained close to the corresponding Λ CDM stellar-mass budget, rather than producing a catastrophic global excess.

The physical reason is simple: bright rare halos are not most baryons. The rare high-mass tail can change substantially while the integrated stellar mass density changes only modestly. This is a required feature of any viable solution to the high-redshift bright-galaxy problem.

7. Roman Forecasts: Breaking the $A_{\text{lock}} - f_{\text{star}}$ Degeneracy

The UVLF alone cannot fully distinguish between a collapse-gate mechanism and an astrophysical brightening mechanism. Both can increase the number of observed bright galaxies.

The degeneracy breaks when we ask not only how many bright galaxies exist, but where they are. A uniform f_{star} boost is local. It changes luminosities, but it does not create new large-scale spatial structure.

By contrast, a spatially modulated registration field, $S_{\text{collapse}}(z, x)$, predicts that bright high-redshift galaxies should form preferentially in coherence-favored regions. Therefore, the Flux/MIP model predicts UVLF enhancement plus excess clustering plus field-to-field variance.

The main default Roman forecast uses a 2400 deg^2 survey area, 25 deg^2 patches, $M_{\text{UV}} < -21$, and redshift bins $10 < z < 12$, $12 < z < 14$, and $14 < z < 16$.

Tier	Area	Purpose
Roman Deep	$\sim 19 \text{ deg}^2$	UVLF calibration
Roman Medium	$\sim 2400 \text{ deg}^2$	clustering and variance test
Roman Full/Wide	$\geq 5000 \text{ deg}^2$	maximal field-to-field variance test

8. Three Spatial Models

To avoid confusing Flux-specific predictions with generic environmental patchiness, we define three spatial models.

Model A: Uniform Astrophysical Control

$$\chi(x) = 0, \quad A_{\text{lock}} = 0$$

This corresponds to a pure f_{star} explanation. It may match the UVLF amplitude, but it should not generate additional coherence-linked clustering.

Model B: Gaussian Random Field Control

$$\chi(x) = G(x)$$

This represents generic environmental patchiness. It tests whether any random large-scale modulation can mimic the signal.

Model C: Flux/MIP Shell Prediction

$$\chi(x) = W_{\chi}(x)$$

where W_{χ} is a shell-like or coherence-domain field descended from the recombination registration transition. This is the distinctive Flux/MIP prediction.

The Roman discriminants are

$$\Delta b = b_{\text{Flux}} - b_{\text{LCDM}}$$

$$\Delta_{\text{var}} = [\text{Var}_{\text{Flux}} - \text{Var}_{\text{fstar}}] / \text{Var}_{\text{fstar}}$$

$$\xi_{\text{patch}}(r) = \langle \delta_N(x) \delta_N(x+r) \rangle$$

At fixed UVLF amplitude, a positive Δb , positive Δ_{var} , and a scale-dependent ξ_{patch} excess would favor the Flux/MIP lock-in model over a uniform f_{star} boost.

9. Relation to S_8

The S_8 tension motivates considering mechanisms that alter structure formation without destroying CMB agreement. However, the present paper does not claim a completed S_8 solution.

The Flux/MIP collapse gate primarily affects nonlinear registration, not linear growth. Therefore, its immediate strength is in the high-redshift rare-halo sector, not in a direct modification of the late-time matter power spectrum.

A full S_8 analysis requires a separate likelihood-level calculation involving weak lensing, galaxy clustering, $f\sigma_8$, and the nonlinear matter power spectrum.

The correct current statement is that the collapse-gate model is designed to remain compatible with late-time growth constraints while addressing the high-redshift bright-tail problem. Potential S_8 relief remains a future extension, not a result established here.

10. Discussion

The Flux/MIP model changes the interpretation of early galaxy formation. In the standard picture, bright $z > 10$ galaxies are explained by a combination of rare halos, high star-formation efficiency, bursty histories, low dust, possible top-heavy stellar populations, or AGN contamination.

In the registration picture, the earliest bright galaxies are not merely statistical outliers. They are the first large-scale structures whose overdensities crossed both a gravitational and informational threshold after recombination.

This reframes the first galaxies as early MIP-locked coherence structures. They are not evidence that gravity became dramatically stronger. They are evidence that the post-recombination universe became better able to preserve bound structure.

The model's strongest feature is its selectivity: rare-tail enhancement without global overproduction. That selectivity is why the model can lift the bright UVLF while remaining compatible with reionization and stellar-mass-density constraints.

11. Conclusion

We have presented a Flux/MIP collapse-gate model in which recombination is interpreted as a transition from a scattering-dominant to a registration-dominant medium. This transition leaves a delayed memory kernel that slightly lowers the effective collapse-registration threshold for rare overdensities.

The resulting mechanism, $\delta_c \rightarrow \delta_{c,eff}$, selectively enhances the high-mass halo tail at $z \geq 10$, providing a physically motivated route to bright JWST galaxies without requiring a broad change to the linear growth factor or the background expansion.

The model remains preliminary, but it makes a clear falsifiable prediction. A uniform astrophysical boost can raise the UVLF, but it should not create distinctive large-scale clustering or extreme field-to-field variance at fixed luminosity. A spatially coherent registration field should.

Future Roman wide-field measurements of $z > 10$ galaxy clustering, patchiness, and scale-dependent variance therefore provide the decisive test.

Appendix A: Registration Rate

$$\Gamma_{loss}(z) = n_e(z) \sigma_T c$$

The loss rate.

$$\Gamma_{reg}(z) = A_{reg} H(z) C_\sigma(z) + B_{reg} R_\sigma(z)$$

The registration rate.

$$I(z_*) = \Gamma_{reg}(z_*) - \Gamma_{loss}(z_*) = 0$$

The transition condition.

$$R_\sigma(z) = (1+z)H(z)|dC_\sigma/dz|$$

The coherence transition shell.

Appendix B: Collapse-Gate Halo Response

$$\delta_{c,eff}(z,x) = \delta_c \{1 - A_{lock} S_0(z)[1 + \epsilon_\chi \chi(x)]\}$$

Effective collapse-registration threshold.

$$v_{eff} = \delta_{c,eff} / \sigma(M,z)$$

Effective peak-height parameter.

$$n_{Flux} / n_{LCDM} \sim \exp[(\delta_c^2 - \delta_{c,eff}^2)/(2\sigma^2(M,z))]$$

Approximate rare-tail abundance response.

Appendix C: Roman Forecast Metrics

$$\Delta_{\text{var}} = [\text{Var}_{\text{Flux}}(\text{N}_{\text{bright}}) - \text{Var}_{\text{fstar}}(\text{N}_{\text{bright}})] / \text{Var}_{\text{fstar}}(\text{N}_{\text{bright}})$$

Field-to-field variance discriminant.

$$\Delta b = b_{\text{Flux}} - b_{\text{LCDM}}$$

Bias discriminant.

$$\xi_{\text{patch}}(r) = \langle \delta_{\text{N}}(x) \delta_{\text{N}}(x+r) \rangle$$

Patch clustering proxy.

References and Data Notes

- Donnan et al. (2024), JWST/NIRCam high-redshift UV luminosity function measurements and early-galaxy abundance analysis.
- Whitler et al. (2025), JADES UV luminosity function constraints and high-redshift dropout samples.
- Roman Space Telescope High-Latitude Wide-Area Survey planning documentation, including survey tiers and depth assumptions.
- This manuscript also refers to internal diagnostic pipeline layers v1-v9: recombination registration, collapse-gate halo response, reionization consistency, stellar-mass-density consistency, and Roman clustering forecasts.